C Programming Syntax Guide

Header Files

- Include directive: #include <filename.h> or #include "filename.h"
- Standard I/O: #include <stdio.h>
- String handling: #include <string.h>
- Mathematical functions: #include <math.h>
- General utilities: #include <stdlib.h>

Structure of C Program

- Variable_name can be any name like a,b,average,sum ,area,temp,c, etc...
- Preprocessor commands (e.g., #include)
- Main function: void main() { /* code */ }
- Void main(){} or int main(){/*code*/ return 0}
- [return is not compulsory can return 1 or any integer like return 100 or return 5 etc.]
- Variable declarations: int variable_name [Example int a,b; or float variable name] ;
- Comments: /* multi-line comment */ or // single-line comment

Escape Sequences

- \a : Alarm or Beep
- \n : New Line
- \t : Horizontal Tab
- \v : Vertical Tab
- \\ : Backslash
- \' : Single Quote
- \" : Double Quote
- \b : Back Space
- \0 : NULL character

Data Types

- int : Integer (4 bytes, e.g., 10)
- float : Floating point (4 bytes, e.g., 10.50)
- double : Double precision float (8 bytes)
- char : Single character (1 byte, e.g., 'a')

Input & Output

- printf(“”,variable_name if any);
- printf(“”,variable_name if any): Outputs data to console
- Example printf(“%d + %d = %d”,a,b,a+b).
- scanf(“”,%variable_name);
- scanf(“format specifier”, &variable_name): Reads data[like radius, numbers or any input that is wanted from user can take multiple input also].
- Common Format Specifiers: %d (int), %f (float), %c (char), %s (string), %lf (double)

Operators

- Arithmetic: +, -, *, /, % (modulo)
- Relational: <, <=, >, >=, ==, !=
- Logical: && (AND), || (OR), ! (NOT)
- Assignment: =, +=, -=, *=, /=, %=
- Increment/Decrement: ++, --
- Conditional (Ternary): exp1 ? exp2 : exp3
- Bitwise: &, |, ^, <<, >>
- Special: sizeof, &, * (pointer), , (comma)

Math Functions (requires <math.h>)

- ceil(number) : Rounds up
- floor(number) : Rounds down
- sqrt(number) : Square root

- pow(base, exponent) : Power
- abs(number) : Absolute value